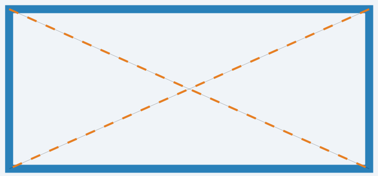
Understanding Area

Measuring Two-Dimensional Space



Mathematics Department

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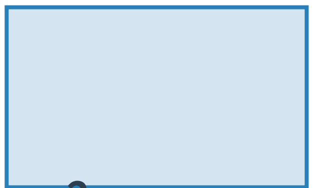

1 $unit^2$ per square

Area of a Rectangle

To find the area, multiply the base by the height.

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Area = b × h
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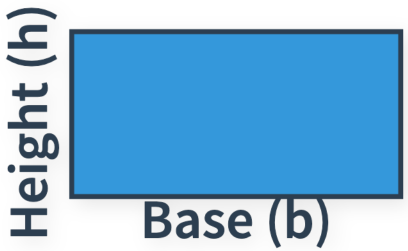

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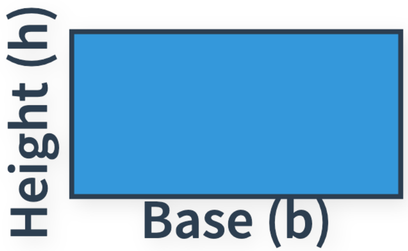
$$\text{Area} = \mathbf{b} \times \mathbf{h}$$

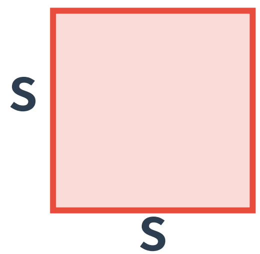
Note: Often expressed as Length × Width.



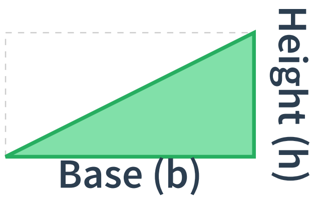
Special Case: The Square

Since all sides are equal in a square (s), the formula simplifies:

$$\text{Area} = s \times s = s^2$$



Area of a **Triangle**



A triangle is exactly **half** of a rectangle with the same base and height.

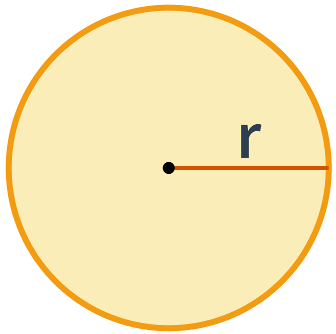
> $\text{Area} = \frac{1}{2} \times \mathbf{b} \times \mathbf{h}$

Area of a Circle

Requires the radius (r) and the constant π (approx. 3.14159).

$$\text{Area} = \pi r^2$$

Definition: **Radius** is the distance from the center to the edge.



Formula Summary

Shape	Formula	Key Variables
Rectangle	$b \times h$	Base, Height
Square	s^2	Side
Triangle	$\frac{1}{2}bh$	Base, Vertical Height
Circle	πr^2	Radius

Quick Check!

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⚠️ Remember: Units must be **squared**!

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LESSON COMPLETE